Abhishek Jain

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Summary

Data Scientist with 7+ years of experience applying Python, SQL, statistical modelling, machine learning, to solve large-scale business problems in data-intensive environments. Experienced in building end-to-end explainable models, translating business problems into analytics solutions, and deploying scalable data products. A strong communicator with a proven track record of driving data-driven decision-making in Agile teams.

Experience

- **Amazon** — Hyderabad, India
  Senior Data Scientist — *Jan 2024 – Present*
  - Deployed Downstream Impact (DSI) model that utilizes the Two-Stage Causal Inference framework to quantify the long-term revenue impact of customer behavior across US, EU5 and JP, delivering actionable insights that drove **over $500MM** in annualized business value. Leveraging **Logistic Regression** for propensity scoring, **Ridge Regression** for causal effect estimation, **PCA** to compress 2000+ feature set and **k-Means Clustering** to create cluster distance features.
  - Productionized a **text classification framework using NLP and XGBoost** for ~50K/month SDS contacts and 10K+ feedback responses, achieving **78% accuracy** on a 12-level taxonomy to surface systematic PARP delivery defects. Scaled the methodology to 5+ delivery streams, including Hub, DSP and Flex drivers.
  - Designed a self-serve Text-to-SQL platform leveraging **GenAI** and a RAG-enabled knowledge base to translate natural language into complex customer targeting queries, achieving **~90% execution accuracy** and reducing ad-hoc analytics requests by ~65%.
  - Analyzed 12M customer interaction and transaction records to test hypotheses around engagement decay, behavioral volatility, and negative experience-driven churn. Engineered 150+ recency, frequency, concessions and stability features and built explainable **Logistic Regression and LightGBM models**, achieving early churn risk detection with a **recall of 85%.** Applied **SHAP-based** explainability to ensure transparent and ethical model behavior.
  - **Partnered with Country Heads** to analyze the impact of shipping cost savings on overall contribution profits (CP), recommending strategic changes in eligibility criteria that drove **~8% incremental orders** resulting in increase of PARP CP by **23% YoY**. Unblocked 5 GL categories that further drove CP by 4% achieving Director Goal.
  - **Collaborated with marketing director** to re-engage customers lost due to decommissions, built a self-serve Flask based tool to find nearest alternative pickup points, driving **~67% of customer retention** across 20 decommissions in EU and FE regions.
  - Leading a team of 4 BIEs. Driving data science and analytics charter of Pickup & Returns vertical within Last Mile org, managing cross-functional delivery commitments and leadership escalations.

- **The Math Company** — Bangalore, India
  Senior Data Scientist — *June 2021 – Dec 2023*
  - Designed and implemented a modular time-series forecasting framework in Databricks with an integrated feature store, forecasting hierarchical demand across region, marketplace, product group, brand, and SKU levels. Reduced experimentation runtime by ~80% and enabled rapid comparison across statistical and ML models (ETS, ARIMA, AutoARIMA).
  - Developed a tire quality prediction and process-optimization tool using XGBoost and a Power BI-based front end, simulating the impact of machine-setting changes on defect rates. Estimated annual client savings of €4M.
  - Built a self-service analytics solution for sealant compound application using high-frequency laser sensor data, applying FFT and convolutional matrix reduction techniques to detect anomalies. Improved operational efficiency by 97%.
  - Created repeat-customer proxy logic using statistical tests and behavioral signals (purchase frequency,

geography, seasonality), enabling leadership to quantify repeat vs new customer demand and improve planning accuracy.

- Developed a commodity price forecasting and decision-support tool to predict natural rubber prices using macroeconomic indicators, supporting hedging strategies. Augmented forecasts with NLP-based sentiment scoring of relevant market news.

- **Wipro** Noida, India
  Analyst *Nov 2019 – Jun 2021*
  - Built an end-to-end rolling 13-month workflow health dashboard in Tableau, from data ingestion and metric definition to deployment, tracking 350+ metrics across 10+ operational domains.
  - Designed and optimized ETL workflows that reduced manual reporting effort by 40% and improved data reliability.
  - Developed a VBA-based quality management tool with version control, increasing automated quality checks by 74% and improving reporting consistency.

- **Nurtr; Thought Roots Pvt. LTD.** Gurugram, India
  Data Analyst *Sep 2018 – Oct 2019*
  - Developed a lead classification machine learning model to predict hot vs cold leads, increasing conversion rates from ~40% to ~70%.
  - Designed and implemented a lightweight CRM solution to support sales and business development teams, contributing to a 20% increase in upselling.
  - Delivered executive-level dashboards and insights (GMV, revenue, collections) to support sales target planning and tracking.

Education

- **Liverpool John Moores University** UK
  Master of Science in Data Science *Mar 2025 – Mar 2026*
- **International Institute of Information Technology** Bangalore, India
  PG Diploma in Data Science *Mar 2020 – Mar 2021*
- **Inderprastha Engineering College** NCR, India
  B. Tech CE *Aug 2013 – Jun 2017*

Skills

- **Programming:** Python, SQL, R
- **Machine Learning & Stats:** Supervised & Unsupervised Learning, XGBoost, CatBoost, Random Forests, Decision Trees, Logistic/Linear Regression (Ridge, Lasso, Elastic Net), K-Means & Hierarchical Clustering, Dimensionality Reduction (PCA), A/B Testing, Time Series Forecasting (ARIMA, SARIMA, SARIMAX, VAR)
- **Deep Learning & NLP:** PyTorch, Neural Networks (CNN, RNN, LSTM, GRU), Transfer Learning, Natural Language Processing (NLP), Anomaly Detection
- **Generative AI & LLMs:** Transformer Models (LLaMA, Mistral, BERT, ViT), RAG (Retrieval-Augmented Generation), PEFT/LoRA Fine-tuning, Model Quantization (FP32 to 8/4-bit), Hugging Face, LangChain, vLLM, Prompt Engineering
- **MLOps & Data Engineering:** Databricks, Flask, Vector Databases (Pinecone, Milvus), Git, Docker, CI/CD
- **Cloud & Visualization:** AWS, Azure, Tableau, Power BI, QuickSight, Data Studio, Excel

AWARDS & ACHIEVEMENTS

- **Mechanism Maven Award 2024** at Amazon for building DSI Causal Inference framework.
- **Appreciation Award 2022** at The Math Company for building tire quality prediction model.
- **Appreciation Award 2021** at The Math Company for turning around client experience.